

6

Building Empirical Models

● **Building Empirical Models** is a systematic process of creating mathematical models based on observed data and experimental results, rather than theoretical physics or chemistry laws.

In engineering, statistics, and system identification, this process typically follows a core **six-step framework**:

1. Experimental Design

- **Goal:** Plan how to collect high-quality, informative data.
- **Method:** Define independent variables (inputs) and dependent variables (outputs). Use Design of Experiments (DoE) like Response Surface Methodology (RSM) to maximize information while minimizing experimental trials.

2. Data Collection / Plant Experimentation

- **Goal:** Execute the designed experiments and record real-world data.
- **Method:** Run trials under controlled conditions. In process control, this often involves applying a specific input signal (like a step change) and recording the dynamic output responses over time.

3. Selection of Model Structure

- **Goal:** Choose the right mathematical form to represent the relationship.
- **Method:** Analyze scatter plots or rely on historical process knowledge. Structures can range from simple linear regressions to First-Order Plus Dead Time (FOPDT) dynamic models or complex non-linear equations.

4. Parameter Estimation

- **Goal:** Calculate the unknown coefficients (e.g., slopes, gains, or time constants) within your chosen model structure.
- **Method:** Apply numerical techniques like the **Ordinary Least Squares (OLS)** method, non-linear regression, or graphical methods (such as the process reaction curve method).

5. Diagnostic Evaluation

- **Goal:** Check if the estimated model meets fundamental statistical assumptions.
- **Method:** Analyze the residuals (errors) to ensure they are randomly distributed with no patterns. Check the Coefficient of Determination (R^2) and evaluate if the model suffers from overfitting or multicollinearity.

6. Model Validation

- **Goal:** Ensure the model can accurately predict future, unseen data.
- **Method:** Test the completed model using a **fresh, independent dataset** that was never used during the parameter estimation step. If the prediction error remains low, the model is ready for deployment.

● **MODELING HYDROGEN PRODUCTION** involves using two- and three-dimensional computer models to enhance extraction from sources like methane and biomass. Researchers at the University of Salerno found that while **3D** models provide more detail on reactant flow, they require significantly more computational power to analyze catalytic plate efficiency.

While methane extraction leaves a carbon footprint, it provides twice the energy of direct combustion; conversely, modeling agricultural biomass reveals potential to reduce atmospheric carbon dioxide. These empirical models are vital for optimizing fuel technology and reducing emissions.

● LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- **1.** Use simple or multiple linear regression to build empirical models from engineering and scientific data.
- **2.** Analyze residuals to check model adequacy and verify underlying statistical assumptions.
- **3.** Test statistical hypotheses and construct confidence intervals for regression model parameters.

- **4.** Use the regression model to estimate a mean response or predict a future observation.
- **5.** Use confidence or prediction intervals to quantify the error in your estimation or prediction.
- **6.** Comment on the specific strengths and weaknesses of your empirical model.

● 6-1 INTRODUCTION TO EMPIRICAL MODELS

An **empirical model** is an equation developed from experimental data rather than from fundamental physical laws.

Key Concepts

- **Data-Driven:** It relies on observed measurements.
- **Approximation:** It approximates system behavior within the tested data range.
- **Prediction:** It forecasts outputs when theoretical equations are too complex.

Role of Regression

Linear regression connects two types of variables:

1. **Dependent Variable (Y)**: The response or output being predicted.
2. **Independent Variables (X)**: The predictors or inputs controlled during testing.

Model Forms

Simple Linear Regression

One predictor variable:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Multiple Linear Regression

Two or more predictor variables:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

- β_i : Unknown parameters estimated from data.
- ϵ : Random error representing measurement noise.

- Engineers use mathematical equations to describe relationships between variables when solving technical problems.

Here is the summary of how these equations are categorized based on their origin:

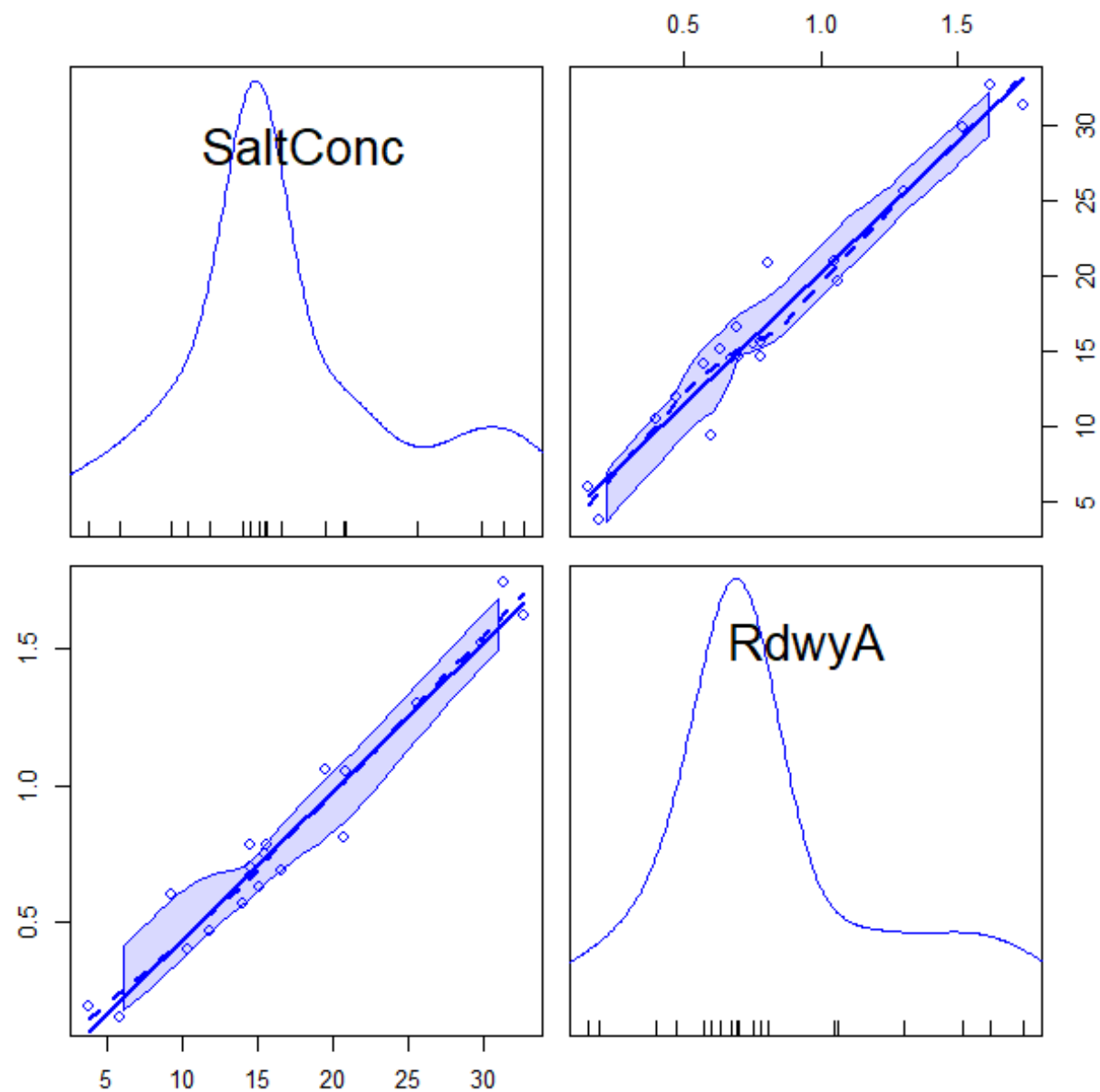
- models**: Equations used by engineers to formulate and solve complex problems.
- mechanistic models**: Equations based on physical, chemical, or engineering science knowledge (e.g., Ohm's law, gas laws, Kirchhoff's laws).

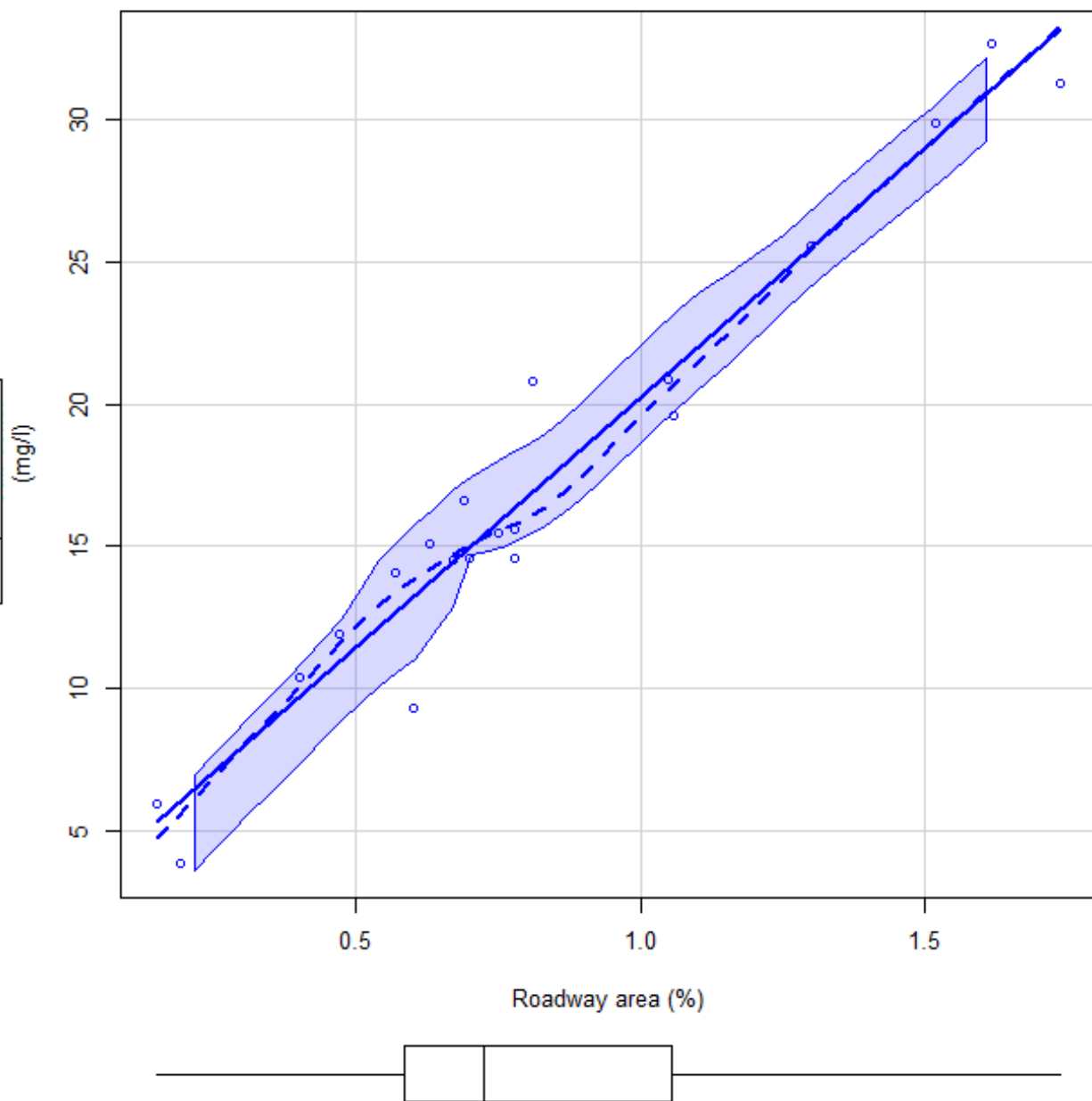
- **empirical model**: An equation built entirely from observed data when the underlying physical mechanism is unknown.

- Data from the **Journal of Environmental Engineering** illustrates a **probabilistic linear model** relating salt concentration in surface streams to paved roadway area, with **Fig. 6-1** showing a positive relationship. The **response variable** (y) and **regressor variable** (x) are modeled via $E(Y|x) = \mu_{Y|x} = \beta_0 + \beta_1 x$ to account for random variation, as detailed in the data below.

Table 6-1 Salt Concentration in Surface Streams and Roadway Area

Observation	Salt Concentration (y)	Roadway Area (x)			
1	3.8	0.19	11	15.6	0.78
2	5.9	0.15	12	20.8	0.81
3	14.1	0.57	13	14.6	0.78
4	10.4	0.40	14	16.6	0.69
5	14.6	0.70	15	25.6	1.30
6	14.5	0.67	16	20.9	1.05
7	15.1	0.63	17	29.9	1.52
8	11.9	0.47	18	19.6	1.06
9	15.5	0.75	19	31.3	1.74
10	9.3	0.60	20	32.7	1.62





- **Simple Linear Regression Model**: A mathematical framework that relates a dependent variable to exactly one independent variable.
- **response**: The dependent variable (Y) that represents the outcome being measured or predicted.
- **regressor variable**: The single independent variable (x) used to explain the variation in the response.
- **regression coefficients**: The model parameters (β_0 and β_1) that define the intercept and slope of the line.

- **(6-1)**: The specific equation number assigned to this **simple linear regression model** formula.

$$Y = \beta_0 + \beta_1 x + \epsilon \quad (6-1)$$

- **Conditional Mean:** When the error term has a conditional expected value of zero given x , the conditional mean of the response variable given x aligns perfectly with the true regression line at any given point x .
- **Conditional Variance:** The conditional variability of the response variable at any specific point x is determined entirely by the constant conditional variance of the error term given x .
- **Slope Interpretation:** The coefficient β_1 represents the exact change in the conditional expected value of the response variable given x for every single-unit increase in the predictor x .

- **Uniform Distribution:** The model assumes a conditional distribution of response values given x exists at each point x , and this conditional distribution given x maintains a constant conditional variance across all levels of the predictor x . 

$$E(Y|x) = E(\beta_0 + \beta_1 x + \epsilon|x) = \beta_0 + \beta_1 x + E(\epsilon|x) = \beta_0 + \beta_1 x$$

$$\begin{aligned} \text{Var}(Y|x) &= \text{Var}(\beta_0 + \beta_1 x + \epsilon|x) \\ &= \text{Var}(\beta_0 + \beta_1 x|x) + \text{Var}(\epsilon|x) = 0 + \sigma^2 = \sigma^2 \end{aligned}$$

- **Graphical Illustration:** **Figure 6-2** visually demonstrates how the conditional distribution behaves along the true regression line.
- **Normality Assumption:** The model uses a conditional normal distribution to represent the random error term.
- **Response Distribution:** The response variable inherits this normality, making it a conditionally normally distributed random variable.
- **Variance Impact:** A small conditional variance keeps observations close to the line, while a large conditional variance causes wider deviations.

- **Homoscedasticity:** The conditional variability remains identical across all levels of the predictor because the conditional variance is constant.

- **Model Application:** The regression model directly defines the statistical relationship between salt concentration (Y) and roadway area (x).
- **Distribution Properties:** For any given roadway area, the salt concentration follows a conditional normal distribution with a conditional mean of $3 + 15x$ and a constant conditional variance of 2 .
- **Point Calculation:** When the roadway area $x = 1.25$, the conditional mean value of the salt concentration is calculated exactly as $\mu_{Y|x} = 3 + 15(1.25) = 21.75$.

- **Variance Consistency:** At this specific point ($x = 1.25$), the conditional variance remains exactly 2, maintaining the constant conditional variance assumption across the model.



- **Multi-Variable Situations:** Many real-world empirical modeling scenarios involve more than one regressor variable.
- **Relationship Description:** A regression model remains an effective tool to mathematically describe these multi-variable relationships.
- **Model Definition:** A regression model that contains more than one regressor variable is called a **multiple regression model**.

- **Tool Life Example:** The model describes how the effective life of a cutting tool (Y) depends on two distinct factors: cutting speed (x_1) and tool angle (x_2).
- **Model Classification:** This framework represents a **multiple linear regression model** specifically featuring two regressors along with a random error term (ϵ).
- **Definition of Linearity:** The term **linear** is applied because the equation forms a linear function of its unknown parameters (β_0 , β_1 , and β_2), rather than the shape of the independent variables themselves.

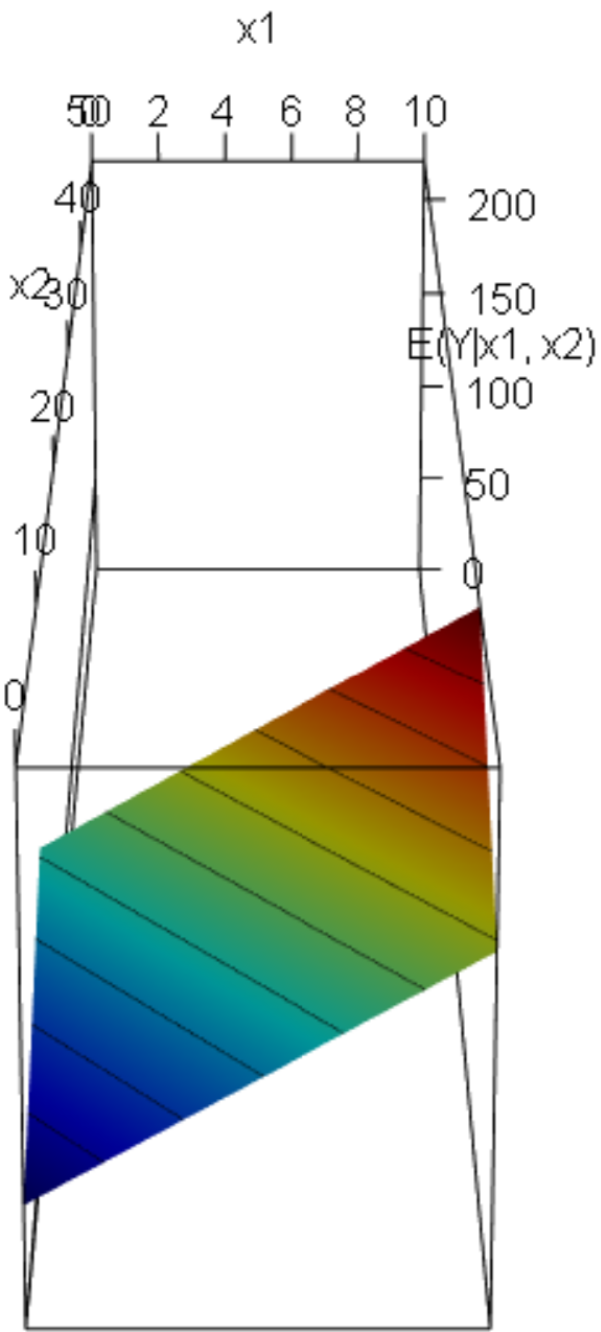
- **Equation Identifier:** The mathematical expression for this relationship is designated by the equation number **(6-2)**.

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \quad (6-2)$$

- **Geometric Representation:** **Figure 6-3** illustrates a three-dimensional regression plane alongside its corresponding **contour plot**.
- **Plane Intercept:** The parameter β_0 represents the **intercept**, which defines the height of the plane where it intersects the response axis.
- **Coefficient Definition:** The parameters β_1 and β_2 are called **partial regression coefficients** because they isolate the effect of each individual regressor.
- **Slope Interpretation:** The coefficient β_1 measures the conditional expected change in Y per unit change in x_1 while keeping x_2 constant, and β_2 behaves conversely.

- **Contour Visualization:** A **contour plot** projects the three-dimensional surface onto a two-dimensional space, displaying straight, parallel lines of constant conditional expected values.

$$E(Y|x_1, x_2) = 50 + 10x_1 + 7x_2$$



- **Multiple Linear Regression Model**: A statistical framework used when a single **dependent variable** is influenced by a set of multiple **independent** variables.
- **response**: The outcome variable (Y) being measured, modeled, or predicted by the system.
- **regressor variables**: The collection of k independent inputs ($x_1, x_2, \dots, x_k$) used to explain variations in the response.
- **hyperplane**: The multi-dimensional geometric surface described by this **multiple linear regression model** within the joint space of all variables.

- **Coefficient Interpretation:** Each regression coefficient β_j represents the conditional expected change in the response Y per unit change in x_j when all other regressors are held constant.
- **(6-3):** The mathematical expression defining this general multi-variable relationship is designated by equation number **(6-3)**.

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \epsilon \quad (6-3)$$

- **Practical Application:** Multiple linear regression structures are frequently utilized as **empirical models** in engineering and research.
- **Mechanism Unknown:** They are chosen precisely when the underlying physical, chemical, or mechanistic model relating the variables remains unknown.
- **Adequate Approximation:** These frameworks serve as a highly effective localized approximation of the system behavior over specific, defined ranges of the independent variables.



- **Mathematical Foundation:** These empirical models are deeply connected to the well-known Taylor series approximation technique used for simplifying complex mathematical functions.
- **First-Order Approximation:** A first-order Taylor series approximation of an unknown function around a sample mean simplifies directly into a simple linear model when the remainder term is omitted.
- **Multi-Variable Expansion:** Complex relationships involving a function with multiple independent variables can be effectively captured using a higher-order Taylor series expansion.

- **Practical Analysis:** Because of this mathematical link, complex and unknown functional relationships can often be successfully analyzed using general multiple linear regression techniques. 

$$f(x) = f(\bar{x}) + f'(\bar{x})(x - \bar{x}) + R \approx \beta_0 + \beta_1 x$$

- **Polynomial Structure:** The initial framework uses a cubic polynomial model containing a single independent variable to capture non-linear trends.
- **Variable Transformation:** Setting new individual terms equal to successive powers of the original variable converts the higher-order terms into separate distinct components.
- **Linear Classification:** The transformed expression represents a standard multiple linear regression model containing three unique regressor variables.

- **Equation Identifiers:** The original cubic expression is marked as **(6-4)**, and its newly transformed linear counterpart is marked as **(6-5)**.

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon \quad (6-4)$$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon \quad (6-5)$$

- **Joint Effects:** Models that incorporate an **interaction** between variables can be analyzed using multiple linear regression techniques.
- **Cross-Product Representation:** An **interaction** between two distinct variables is represented by adding a cross-product term into the equation.
- **Linear Transformation:** Defining a new variable as the cross-product of the original variables transforms the interaction expression into a standard linear structure.

- **Equation Identifier:** This specific relationship is identified by the equation number **(6-6)**.

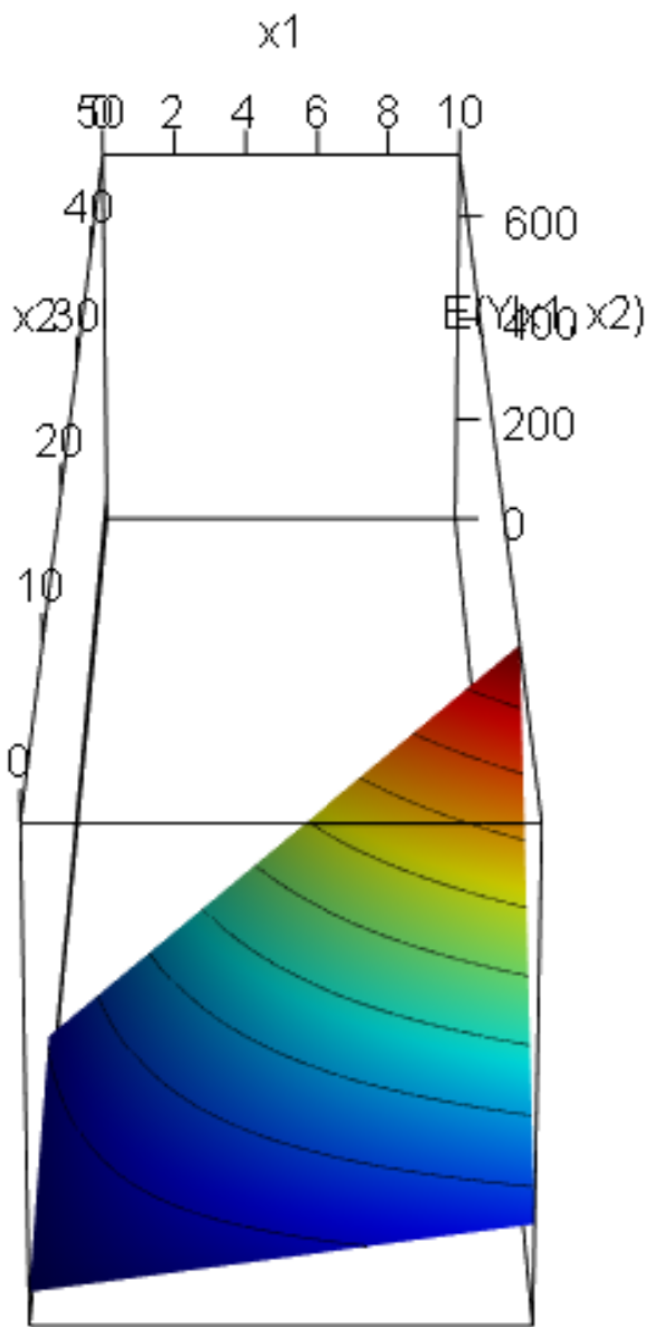
$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \epsilon \quad (6-6)$$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

- **Visual Representation:** **Figure 6-4** presents both a three-dimensional plot and its two-dimensional counterpart to show a model containing a cross-product term.
- **Surface Curvature:** The addition of this cross-product element causes the model to generate a non-linear geometric shape rather than a flat plane.

- **Core Classification:** The fundamental rule of this methodology dictates that **any regression model** that is **linear in parameters** (the β coefficients) is a **linear regression model**, **regardless** of the **shape** of the **surface** that it **generates**.

$$Y = 50 + 10x_1 + 7x_2 + 5x_1x_2$$



- **Graphical Meaning:** The provided figures offer a clear visual interpretation of how independent variables combine to influence an outcome.
- **Core Definition:** An interaction implies that the specific effect produced by changing one variable depends directly on the current level of another variable.
- **Numerical Demonstration:** Changing x_1 from 2 to 8 produces a much smaller change in the conditional expected response when $x_2 = 2$ than when $x_2 = 10$.

- **Engineering Relevance:** Interaction effects occur frequently in product design, process optimization, and other engineering activities.
- **Analytical Method:** Regression analysis serves as a primary technical tool used by engineers to mathematically describe and evaluate these joint behaviors.

- **Model Structure:** The example presents a second-order polynomial model that includes quadratic terms and a cross-product term for interaction.
- **Linear Transformation:** By defining new variables for the higher-order terms, this expression is easily rewritten as a standard multiple linear regression model.
- **Visual Representation:** **Figure 6-5** displays the three-dimensional surface and two-dimensional contour lines generated by the specific numerical model.

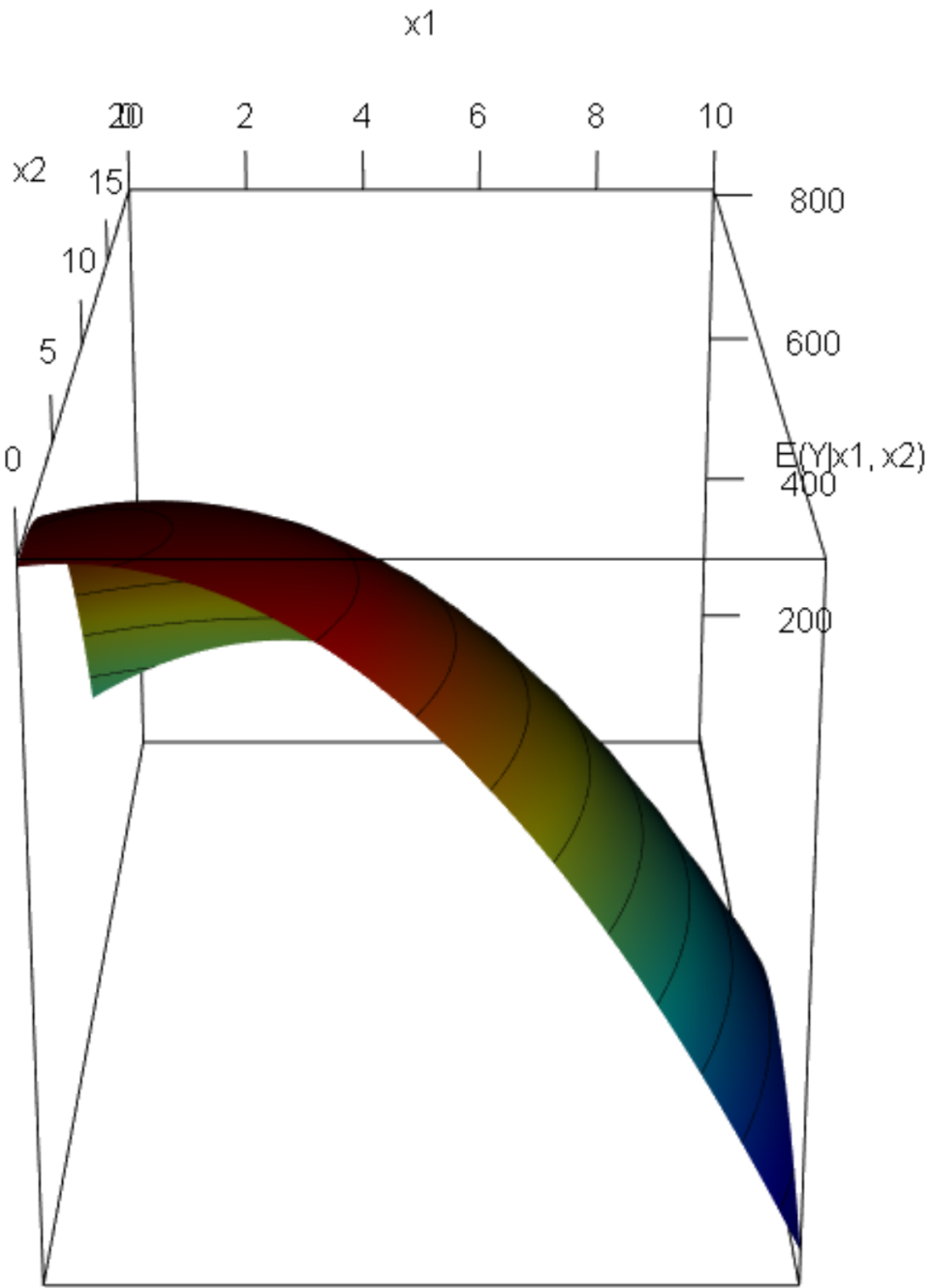
- **Surface Shape:** The combination of quadratic and interaction elements forms a distinct mound-shaped function.
- **System Dynamics:** The plots reveal that the conditional expected change in Y when x_1 is modified by one unit is a function of both x_1 and x_2 .
- **Model Flexibility:** This framework is highly adaptable and can assume a wide variety of geometric shapes depending on its calculated coefficients.

- **Equation Identifier:** This foundational second-order relationship is designated by the equation number **(6-7)**.

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \epsilon \quad (6-7)$$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \epsilon$$

$$E(Y|x_1, x_2) = 800 + 10x_1 + 7x_2 - 8.5x_1^2 - 5x_2^2 + 4x_1x_2$$



- **Parameter Estimation:** In real-world problems, the true values of the regression coefficients (β_j) and the error variance (σ^2) are unknown and must be calculated using sample data.
- **Core Definition:** **Regression analysis** is a collection of statistical tools designed specifically for finding estimates of these unknown parameters in a regression model.
- **Model Application:** Once the parameters are estimated, the resulting fitted equation is typically used to predict future individual observations of Y or to estimate the conditional mean response at a specific level of x .

- **Practical Illustration:** For example, an environmental engineer can use this method to estimate the mean salt concentration in surface streams when the paved road area is exactly $x = 1.25\%$.